

Compositional Scene-Motion Learning via Object-Centric Neural Representations

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Abstract—Compositionality enables scalable and data-efficient robot learning by representing complex behavior as combinations of simpler elements. Building on this principle, we propose a learning-from-demonstration framework that connects perception and motion through smooth object-level representations, enabling motion generation across variations in object configurations. Scenes are modeled using a spatial mixture-of-experts (MoE) of object-centric neural fields with latent-conditioned deformations, while motion is generated through a temporal MoE that combines object-conditioned movement primitives over time. This spatiotemporal compositionality preserves the efficiency of movement primitives while grounding motion in scene structure. Experiments in simulation and on real robots demonstrate strong generalization, data efficiency, robustness to scene variations, and compatibility with both language-based segmentation and 3D scene representations.

I. INTRODUCTION

Learning from Demonstration (LfD) [1], [2] is a widely used approach for teaching robots task-specific skills from human demonstrations. A central challenge is learning diverse, long-horizon behaviors from limited data. Compositionality offers a natural way to address this challenge by decomposing tasks into simpler, reusable components. Such formulations improve data efficiency and generalization compared to monolithic approaches [3]. Prior work has explored compositional motion representations, for example, through funnel-based approaches to guide sequential behaviors [4]. Motivated by these observations, we consider compositionality in both scene and motion representations, where different objects can influence different parts of a manipulation task (Fig. 1).

Movement primitives (MPs) [5]–[9] are a standard representation for encoding and generalizing demonstrated skills. They provide smooth, compact trajectory models that can be learned from few demonstrations. MPs are often used compositionally to represent atomic skills, which can then be combined into more complex behaviors. However, this compositional structure is typically imposed manually, for example by concatenating primitives or specifying transitions through via-points. Moreover, generalization is commonly achieved through hand-designed, low-dimensional task parameters, such as object or goal positions. Identifying suitable parameters and estimating them consistently across task

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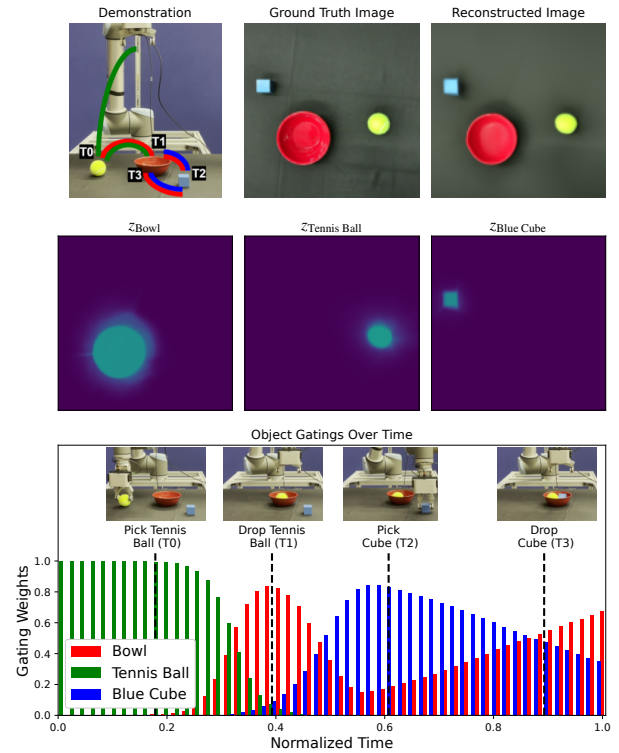


Fig. 1. System overview. Top: Task sequence with object interactions, followed by the ground-truth and reconstructed images. Middle: Object-specific masks inferred from latent codes, illustrating object-centric scene decomposition. Bottom: Temporal gating weights showing each object’s influence over the trajectory; snapshots above the plot correspond to key events (T0–T3), marked by vertical dotted lines.

variations can be challenging. Image-conditioned alternatives [7] can instead learn task representations directly from observations, reducing feature engineering, but typically require substantially more data and sacrifice some of the data-efficiency advantage of movement primitives.

Neural field movement primitives (NFMP) [8] extend MPs by jointly modeling scenes and motions through a shared latent representation, allowing motions to generalize directly with variations in the observed scene. However, NFMP represents the entire scene with a single latent code, making it increasingly difficult to model tasks as the number of objects or task variations grows. To address this limitation, we introduce compositional structure into both scene and motion modeling. Rather than representing the scene as a single entity, we learn smooth object-level latent representa-

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tions that can be selectively incorporated into motion generation. Scenes are modeled using a spatial Mixture-of-Experts (MoE) with object-centric neural fields, while motion is generated through a temporal MoE that combines object-conditioned movement primitives over time. Object-centric structure is encouraged through soft mask-based importance sampling, allowing each expert to specialize without requiring precise object segmentation. Temporal gating enables multiple object representations to contribute within the same motion segment, capturing the relational nature of manipulation. Figure 1 illustrates this spatiotemporal decomposition, showing the object-centric scene representations and their changing influence over the motion.

II. METHOD

We consider learning robot motion from demonstration in settings where task variation is induced by changes in object configurations. Each demonstration consists of a scene observation, coarse object masks used only during training to roughly localize objects, and a corresponding time-aligned trajectory. The demonstrations span the range of object configurations considered for the task, while the trajectories share an approximate temporal alignment across demonstrations. Our goal is to learn smooth object-centric latent representations that capture variations of individual objects and use these representations to generate the corresponding motion for novel scene configurations.

A. Compositional Object-Centric Scene Representation

We represent a scene compositionally using a set of N object-specific neural fields. Rather than encoding the entire scene through a single latent representation, each object i is associated with its own latent vector z_i , allowing variations of individual objects to be represented independently. Each object is modeled by a canonical appearance field F_i together with a deformation field Δ_i conditioned on z_i through Feature-wise Linear Modulation (FiLM) [10], while a separate field F_{bg} represents the static background.

We model the complete scene using a spatial Mixture-of-Experts (MoE), where $\hat{I}$ denotes the reconstructed image, x is a spatial image coordinate, and M_i and M_{bg} denote the object-specific and background masks, respectively:

$$\hat{I}(x|\{z_i\}_{i=1}^N) = \sum_{i=1}^N M_i(x_i) F_i(x_i) + M_{bg} F_{bg}(x), \quad (1)$$

where $x_i = x - \Delta_i(x | z_i)$ denotes the deformed coordinate for object i . The appearance fields F_i model objects in their canonical configurations, while $\Delta_i(x | z_i)$ captures their variation across demonstrations through spatial deformation. The masks are learned through a spatial softmax over the object and background components, allowing the individual fields to be composed into the complete scene.

This formulation associates each object with a continuous latent representation of its observed variations: changes in the latent representation produce smooth changes in the corresponding deformation field, while variations of different objects remain separated across object-specific latent spaces.

We further encourage this smoothness using Lipschitz regularization [11] on the latent-to-deformation mapping.

During training, coarse object masks guide the specialization of the spatial experts without requiring precise object segmentation. For each object, sampling is concentrated around its coarse mask, while pixels belonging to other objects are replaced with samples from the learned background model. This prevents individual experts from explaining variations associated with other objects and encourages scene variations to be decomposed into their corresponding object-specific representations. We additionally use a position-aware zoom-based curriculum around each object during training, improving the representation of small objects by increasing their effective pixel coverage.

B. Latent Relabeling

The object embeddings learned during scene reconstruction are initially unconstrained and do not necessarily provide a consistent parameterization of the observed object variations. To obtain a compact and structured latent representation, we relabel these embeddings using a small set of endpoint latent vectors that span the demonstrated variations. For each object i , these endpoints form a corner set C^i , and demonstration-specific representations are expressed as convex combinations of its elements:

$$\hat{z}_i^{(k)} = \sum_{j=1}^{|C^i|} \alpha_j c_{i,j}, \quad \alpha_j \geq 0, \quad \sum_j \alpha_j = 1, \quad (2)$$

where $c_{i,j} \in C^i$ and α_j are the interpolation weights.

The corner set is constructed iteratively, starting from embeddings corresponding to distant object configurations and expanding the set when the current convex region cannot accurately reconstruct a training instance. This process continues until the demonstrated configurations can be represented from the resulting corner set. Intermediate configurations can then be represented through interpolation within this latent region, providing a structured representation of the variations observed for each object. At inference time, object representations for a new scene are recovered by searching within the same region, naturally constraining generalization to variations supported by the demonstrations.

C. Compositional Motion Generation

The object-centric latent representations provide task parameters for motion generation, allowing changes in individual object configurations to induce corresponding variations in the generated motion. Given the relabeled object representations $\{\hat{z}_i\}_{i=1}^N$, we assume sequential compositionality: different parts of a trajectory can depend on different subsets of objects. We model this through a time-dependent conditioning signal

$$c(t) = \sum_{i=1}^N w_i(t) c_i(\hat{z}_i), \quad w_i(t) = \frac{\exp(s_i(t))}{\sum_{j=1}^N \exp(s_j(t))}, \quad (3)$$

where $s_i(t)$ are learnable relevance scores and $c_i(\hat{z}_i)$ denotes the conditioning generated from the representation of object

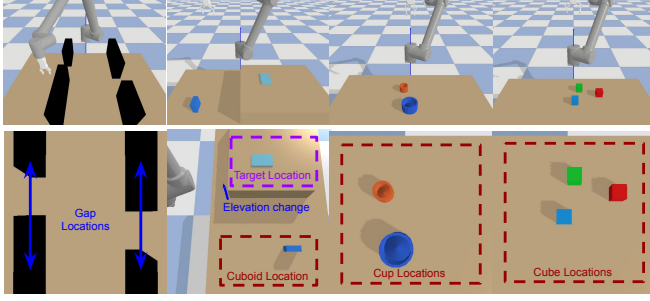


Fig. 2. Simulation tasks with corresponding scene variations. The bottom row shows the camera images used by the methods. In these images, arrows highlight geometric changes and rectangles mark possible object locations.

i. The temporal weights allow different objects to influence different phases of the trajectory, while multiple objects can contribute simultaneously when a motion depends on their relationship.

The resulting conditioning signal is used by a trajectory generator to predict the robot motion over time. In our implementation, $c_i(\hat{z}_i)$ corresponds to FiLM modulation parameters generated from each object representation and combined according to $w_i(t)$. This provides a continuous connection between variations in the learned object representations and variations in the generated motion, while the temporal mixture determines which object representations are relevant throughout the task.

For orientation trajectories, we follow [12] and represent quaternions in axis-angle form in the tangent space of a reference quaternion, yielding continuous rotation trajectories that can be modeled together with Cartesian motion.

D. Inference

At inference time, object-specific latent representations are recovered from a new scene by searching over convex combinations of the learned corner sets and refining the resulting candidates through reconstruction error minimization. This constrains inference to the region of object variations represented by the demonstrations. The recovered object latents are then used directly by the compositional motion model to generate the corresponding trajectory.

III. EXPERIMENTS

We evaluate the proposed framework in simulation and on real-robot manipulation tasks. The simulation studies assess generalization and data efficiency across tasks with increasing numbers of objects, while the real-world experiments demonstrate robustness to perception noise, category-level generalization, and compatibility with both 2D and 3D scene representations.

A. Simulation Experiments

We evaluate the method on four simulated manipulation tasks of increasing complexity: Wall Avoidance, Incline Pick-and-Place, Cup Stacking, and Cube Stacking (Fig. 2), spanning variations in object configurations and task parameters.

TABLE I
SIMULATION RESULTS (MED/ACCURACY).

Method (data regime)	Wall Avoidance	Incline Pick-and-Place	Cup Stacking	Cube Stacking
CNN-CNMP (Low)	2.72±0.08/0.67±0.04	4.35±0.27/0.11±0.03	10.85±1.39/0.02±0.01	14.22±0.59/0.00±0.00
CNN-CNMP (High)	1.70±0.26/0.95±0.02	2.39±0.43/0.51±0.15	3.08±0.27/0.16±0.07	6.07±0.32/0.00±0.00
DP (Low)	2.31±0.04/0.79±0.02	3.60±0.08/0.25±0.01	11.13±1.29/0.02±0.02	11.55±0.19/0.00±0.00
DP (High)	0.55±0.03/0.99±0.00	1.49±0.14/0.86±0.04	2.72±0.09/0.26±0.04	4.77±0.61/0.11±0.03
NFMP (Grid)	0.85±0.18/0.98±0.01	NA	NA	NA
DINO-Film (Low)	4.46±0.19/0.65±0.02	5.03±0.13/0.21±0.00	9.81±0.34/0.00±0.00	15.10±0.07/0.00±0.00
DINO-Film (High)	2.27±0.12/0.84±0.00	3.71±0.06/0.31±0.06	6.69±0.06/0.01±0.00	10.48±0.08/0.00±0.00
CNN-Film (Low)	1.41±0.19/0.89±0.02	3.01±0.55/0.40±0.13	8.74±1.83/0.03±0.01	9.26±0.74/0.00±0.00
CNN-Film (High)	0.46±0.09/0.99±0.00	1.18±0.09/0.92±0.05	2.11±0.20/0.26±0.06	1.32±0.03/0.66±0.10
Ours w/o gating (Low)	0.71±0.04/0.99±0.00	0.89±0.03/0.99±0.01	1.10±0.03/0.86±0.05	1.33±0.05/0.85±0.06
Ours (Low)	0.51±0.03/0.99±0.00	0.71±0.02/1.00±0.00	0.65±0.04/0.99±0.00	0.72±0.05/0.97±0.02

Low/High training sizes per task: Wall Avoidance (10/30), Incline Pick-and-Place (20/50), Cup Stacking (20/50), and Cube Stacking (30/100). Bold numbers indicate the best performance in each task under the low- and all-data regimes, respectively. Lower MED and higher accuracy indicate better performance.

We compare against image-conditioned CNMP [7] and Diffusion Policy (DP) [13], as well as CNN-FiLM and DINO-FiLM [14] ablations that replace the proposed object-centric representation with global image features, using an end-to-end trained CNN and a frozen pretrained DINO encoder, respectively. We additionally evaluate our method without temporal gating (Ours w/o gating) and a modified NFMP [8] using FiLM conditioning. NFMP is evaluated only on Wall Avoidance, as its grid-based training becomes intractable with increasing numbers of task parameters.

Performance is evaluated using mean Euclidean distance (MED) and task success rate under different training-set sizes. All models are evaluated over three random seeds. Baselines are evaluated in low- and high-data regimes, while our method uses only the low-data regime. As shown in Table I, our method consistently outperforms the baselines in the low-data regime and remains competitive with, or better than, baselines trained with substantially more demonstrations. CNN-FiLM generally performs better than CNN-CNMP and DP, suggesting that FiLM-conditioned trajectory generation provides a strong inductive bias, but substantially more data is required to approach the proposed method. Comparing CNN-FiLM with our method without temporal gating highlights the benefit of the object-centric representation, while the further improvement from temporal gating demonstrates the importance of selectively incorporating object information throughout the motion. These advantages become more pronounced in the more complex multi-object tasks. The only exception is Wall Avoidance, where high-data CNN-FiLM achieves marginally lower MED, while both methods successfully complete the task.

B. Real-Robot Experiments

We further evaluate the framework on four real-world manipulation tasks (Fig. 3), assessing robustness to perception noise and clutter, category-level generalization, language-conditioned object selection, and compatibility with 3D scene representations. Demonstrations are collected via kinesthetic teaching, except for the final task, where motion planning is used, and trajectories are temporally aligned using gripper-state transitions.

In the first task, the robot picks up tabletop objects and

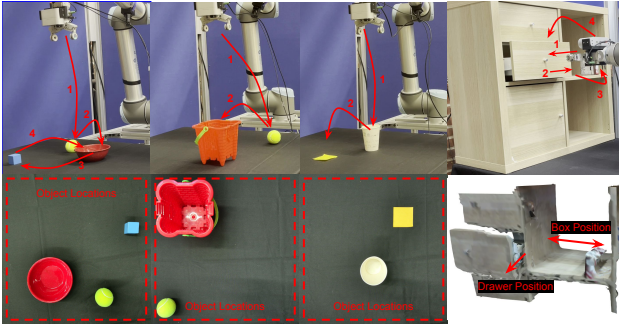


Fig. 3. Real-world tasks and model inputs. For the 3D task, a representative mesh is visualized.

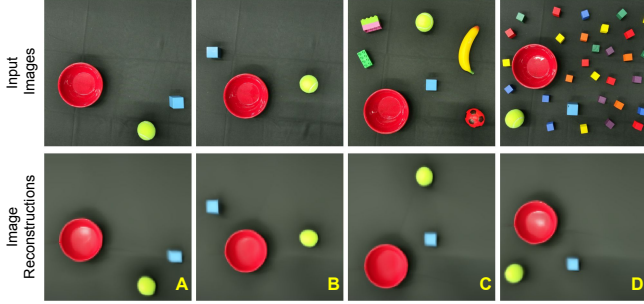


Fig. 4. Input images and corresponding scene reconstructions for the first task, without (A, B) and with (C, D) distractor objects.

places them in a bowl, with object locations varying across trials. Using 30 demonstrations, the system successfully completes all 25 test trials. We further introduce visually distinct distractor objects in five additional trials, all of which are completed successfully. Figure 4 shows representative input images and scene reconstructions with and without distractors.

The second and third tasks evaluate category-level generalization using masks generated by a vision-language segmentation model [15]. In the second task, the robot picks up a ball and places it in a container; in the third, it picks up a cup and places it at a target location marked by a piece of paper. The models are trained with 20 and 30 demonstrations, respectively, and evaluated on both training objects and unseen instances from the same semantic categories. The system achieves 14/15 and 15/15 successful trials, respectively. We additionally evaluate a multi-object, language-conditioned setting, where a user prompt specifies the target object in a cluttered scene. The prompt is used to obtain the corresponding segmentation mask and colorized input, from which the model generates the trajectory. Across all tested trials and prompt variations, the system successfully completes all executions (Fig. 5).

Finally, we evaluate the extension to 3D scene representations on a drawer manipulation task, where a carton box must be placed inside a drawer under varying drawer openings and object positions. Scene representations are constructed using occupancy fields from multi-view scans [16], allowing variations in both the drawer and box configurations to be

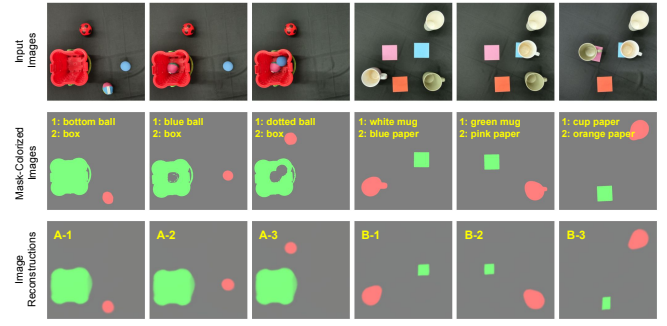


Fig. 5. Language-conditioned manipulation examples. Prompts specify the target object, enabling task execution in cluttered multi-object scenes.



Fig. 6. Input mesh and reconstructed surface (A), and corresponding robot executions at key grasp poses (B, C). The reconstructed surface (shown in blue) captures drawer opening and box position correctly.

represented compositionally in 3D. The model is trained on four demonstrations and evaluated on 10 novel configurations, successfully completing all trials. Figure 6 shows representative scene reconstructions and corresponding robot executions.

IV. CONCLUSION

This paper presents a compositional framework for motion generation from demonstrations that couples object-centric scene representations with temporally structured motion modeling. The approach learns smooth, object-level latent representations from a small number of demonstrations and uses them to generate trajectories through time-varying expert aggregation. Experiments in simulation and real-world settings demonstrate strong data efficiency and generalization across scene variations, highlighting the benefits of compositionality in both scene and motion representations. Despite these advantages, the current formulation relies on coarse object masks during training to encourage object-centric specialization, and performance can degrade in cluttered scenes with visually similar distractor objects. Future work will investigate reducing these supervision requirements and incorporating richer semantic representations to improve robustness and broader scene-level generalization.

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